



Sydney Girls High School
2012
Trial Higher School Certificate
Examination

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11 – 14

This is a trial paper ONLY. It does not necessarily reflect the format or the contents of the 2012 HSC Examination Paper in this subject.

Total marks – 70

SECTION 1 – Pages 2 - 5

10 marks

- Attempt questions 1 – 10
- Allow about 15 minutes for this section

SECTION II – Pages 6 - 9

60 marks

- Attempt questions 11 – 14
- Allow about 1 hours 45 minutes for this section

Name: _____

Teacher: _____

Section I - Total Marks 10

Attempt Questions 1 – 10

Allow about 15 minutes for this section.

(1) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{4x}$

(a) $\frac{4}{3}$

(b) 1

(c) $\frac{3}{4}$

(d) 0

(2) $\frac{d}{dx}[\sin(\log x)] =$

(a) $\cos(\log x)$

(b) $\frac{\cos(\log x)}{x}$

(c) $\frac{\sin(\log x)}{x}$

(d) $-\cos(\log x)$

(3) We can express $\sin x$ and $\cos x$ in terms of $\tan \frac{x}{2}$, for all values of x except

(a) $x = 2\pi, 6\pi, 8\pi, \dots$

(b) $x = \pi, 3\pi, 5\pi, \dots$

(c) $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$

(d) $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \dots$

(4) Which of the following is an expression for $\int \cos^2 8x dx$

(a) $\frac{x}{2} - \frac{\sin 8x}{32} + c$

(b) $\frac{x}{2} + \frac{\sin 8x}{32} + c$

(c) $\frac{x}{2} - \frac{\sin 16x}{32} + c$

(d) $\frac{x}{2} + \frac{\sin 16x}{32} + c$

(5) Which of the following is the correct expression for $\int \frac{dx}{\sqrt{36-4x^2}}$

(a) $\frac{1}{2} \sin^{-1} \frac{x}{6}$

(b) $\frac{1}{2} \sin^{-1} \frac{x}{3}$

(c) $\frac{1}{4} \sin^{-1} \frac{x}{6}$

(d) $\frac{1}{6} \sin^{-1} \frac{x}{3}$

(6) The velocity, v metres per second, of a particle moving in a simple harmonic motion along the x -axis is given by the equation $v^2 = 64 - 16x^2$
What is the period, in seconds of the motion of the particle?

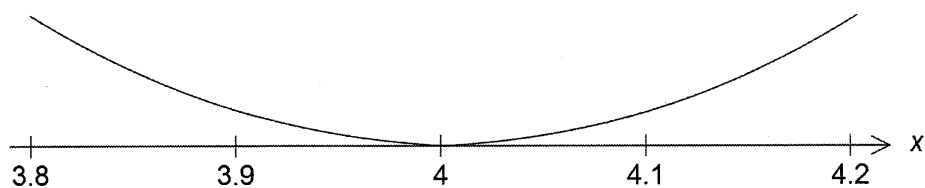
(a) $\frac{\pi}{8}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) π

(7) Part of the graph of $y = P(x)$, where $P(x)$ is a polynomial of degree three, is shown below.



Which of the following could be the polynomial $P(x)$?

- (a) $(x - 4)^3$
- (b) $(x - 5)(x + 4)^2$
- (c) $(x - 1)(x - 4)^2$
- (d) $(x - 1)(x + 2)(x - 4)$

(8) The radius of a sphere is increasing at a rate of 5 centimetres per minute.

What is the rate of increase of the surface area of the sphere, in cubic centimetres per minute, when the radius is 4 centimetres?

- (a) 32π
- (b) 64π
- (c) 100π
- (d) 160π

(9) Which of the following represents the inverse function of $f(x) = \frac{5}{2x-6} - 2$

- (a) $f^{-1}(x) = \frac{5}{2x+4} + 3$
- (b) $f^{-1}(x) = \frac{5}{2x+4} - 3$
- (c) $f^{-1}(x) = 3 - \frac{5}{2x+4}$
- (d) $f^{-1}(x) = \frac{5}{x+2} + 6$

- (10) How many solutions does the equation $\sin 2\theta = \cos \theta$ have in the domain $0 \leq \theta \leq 2\pi$?
- (a) 4
 - (b) 3
 - (c) 2
 - (d) 1

END OF SECTION I

Section II - Total Marks 60

Attempt Questions 11 – 14

Allow about 1 hour 45 minutes for this section

Answer all questions, starting each question on a new sheet of paper

Question 11 (15 marks)

Marks

- (a) Find the acute angle between the lines (to the nearest degree)

2

$$3x + 2y - 6 = 0 \quad \text{and} \quad 2x - y + 8 = 0$$

- (b) A curve has parametric equations $x = \frac{t}{3}, y = 2t^2$.

2

Find the Cartesian equation of this curve.

- (c) Find $\int 2x\sqrt{x-5} \, dx$ using the substitution $u = x-5$

3

- (d) If α, β and γ are the roots of the polynomial $5x^3 - 2x - 4 = 0$, find

2

the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

- (e) Find the exact value of $\tan^{-1}(-\sqrt{3})$.

1

- (f) Find the coordinates of the point P that divides the interval AB externally in the ratio $3 : 2$, where the coordinates of A and B are respectively $(-2, 4)$ and $(3, -6)$.

2

- (g) Solve $\frac{x}{2-x} \geq 2$.

3

Question 12 (15 marks) - Start a new page

Marks

(a) Prove by induction, that $5^n > 20n - 1$ for $n \geq 1$, where n is an integer. 3

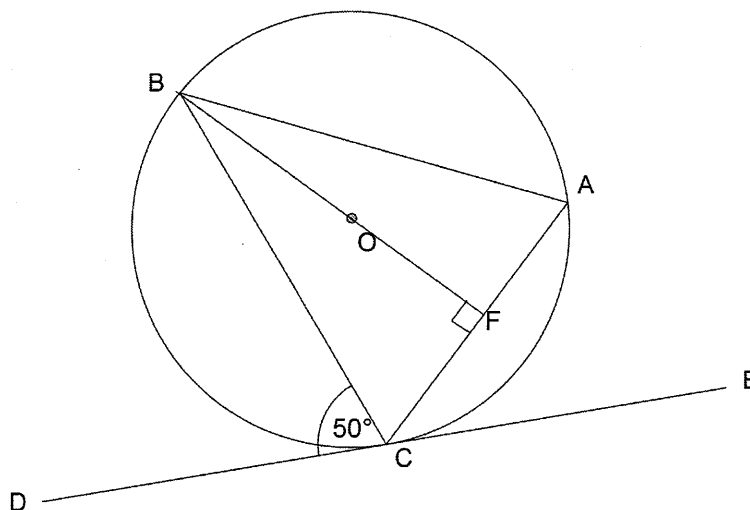
(b) The probability that it rains on any particular day in London is $\frac{2}{3}$.

(i) What is the probability that it does not rain for a whole week in London? 2

(ii) What is the probability that it will rain on only two days during a whole week in London and that these two days are consecutive? 2

(c) Evaluate $\int_0^{\frac{\pi}{2}} \cos x \sin^2 x \, dx$ 2

(d) The line DE is tangent to the circle at C . If $DCB = 50^\circ$, find the size of ACE giving full reasons. 2



(e) Sketch the graph of $y = \sin^{-1}(x - 2)$ 2

(f) The function $f(x) = \sin x - \frac{x}{2}$ has a zero near $x = 2$ 2

Taking $x = 2$ as a first approximation, use one application of Newton's method to find a second approximation to the zero. Give your answer correct to two decimal places.

Question 13 (15 marks) Start a new page**Marks**

(a) Find the exact value of $\int_0^2 \frac{dx}{4x^2 + 20}$ 2

(b) An iron is cooling in a room of constant temperature 20°C . At time t minutes its temperature T decreases according to the equation $\frac{dT}{dt} = -k(T - 20)$ where k is a positive constant.

The initial temperature of the iron is 100°C and it cools to 70°C after 15 minutes.

(i) Verify that $T = 20 + Ae^{-kt}$ is a solution of this equation, where A is a constant. 1

(ii) Find the values of A and k . 2

(iii) How long will it take for the temperature of the iron to cool to 25°C ? 2

(c) Calculate the exact volume generated by the solid formed when $y = \cos^{-1}x$ is rotated about the y -axis between $y = 0$ and $y = \pi$. 3

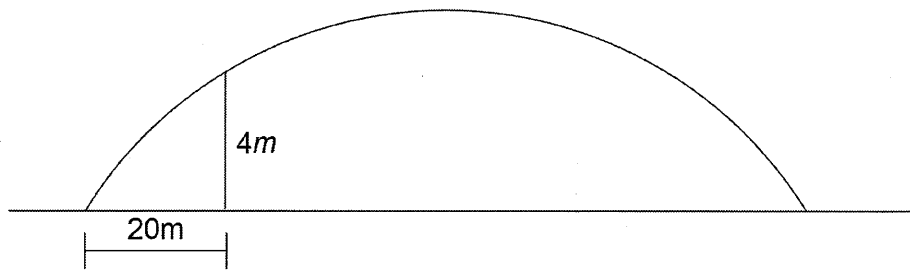
(d) $P(x) = x^4 + 5x^3 + 4x^2 - 8x - 8$

i. Show that $(x + 1)(x + 2)$ is a factor of $P(x)$ 1

ii. Find $Q(x)$ if $P(x) = (x + 1)(x + 2)Q(x)$ 1

(e) Solve the equation $\tan \theta = \sin 2\theta, 0 \leq \theta \leq 2\pi$ 3

(a)



A projectile is fired with initial velocity $V \text{ ms}^{-1}$ at an angle of θ from a point O on horizontal ground. After 5 seconds it just passes over a 4m high wall that is 20 metres from the point of projection. Assume the acceleration due to gravity is 10 m/s^2 . Assume the equations of displacement are $x = Vt \cos \theta$ and $y = Vt \sin \theta - 5t^2$.

- | | | |
|-------|--|---|
| (i) | Find V and θ | 2 |
| (ii) | Find the time taken for the projectile to attain its maximum height. | 1 |
| (iii) | Find the range of the projectile. | 1 |

(b) The points $P(2ap, ap^2)$ and $Q(aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

- | | | |
|-------|--|---|
| (i) | Show that the equation of the chord PQ is $(p+q)x - 2y - 2apq = 0$ | 2 |
| (ii) | Show that the gradient of the tangent at P is p . | 2 |
| (iii) | Prove that if the tangent at P is parallel to the normal at Q then PQ passes through the focus S | 2 |

(c) A particle moves in a straight line so that its acceleration is given by $a = x - 2$, where x is its displacement from the origin.

Initially, the particle is at the origin and has velocity $v = 2$

- | | | |
|------|---------------------------------|---|
| (i) | Find the initial acceleration. | 1 |
| (ii) | Show that $v^2 = (x - 2)^2$ | 2 |
| (iv) | Find x as a function of t . | 2 |

--End of Exam--

Ext 1, Trial 2012
 (1) C $\lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{4}$

(2) B $\cos(\log x) \times \frac{1}{x}$

(3) B $\frac{\pi}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$

(4) D $\cos 16x = \cos^2 8x - \sin^2 8x$
 $= \cos^2 8x - (1 - \cos^2 8x)$
 $= 2\cos^2 8x - 1$

$\frac{\cos 16x + 1}{2} = \cos^2 8x$

(5) B $\frac{1}{2} \int \frac{dx}{\sqrt{9-x^2}}$

(6) C $\frac{1}{2} V^2 = 32 - 8x^2$
 $\ddot{x} = -16x$
 $\omega^2 = 16$
 $\omega = 4$
 $T = \frac{2\pi}{\omega}$

(7) C $x=4$ is a double root

(8) D $S = \pi r^2$
 $\frac{dS}{dt} = 8\pi r$
 $\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt}$
 $= 8\pi r \times 4\sqrt{5}$

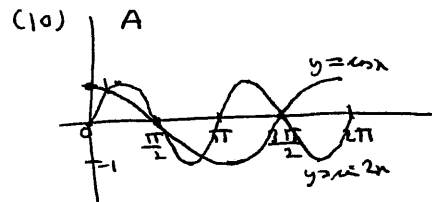
(9) A $x = \frac{5}{2y-6} - 2$

$x+2 = \frac{5}{2y-6}$

$2y-6 = \frac{5}{x+2}$

$2y = \frac{5}{x+2} + 6$

$y = \frac{5}{2x+4} + 3$



Ext 1 2012

11
 a) $\tan \theta = \left| \frac{-\frac{3}{2} - 2}{1 + (-\frac{3}{2} \times 2)} \right|$

$\theta = 60^\circ$ (2)

b) $t = 3x$

$y = 2(3x)^2$
 $= 2(9x^2)$

$y = 18x^2$ (2)

c) $u = x - 5$

$\frac{du}{dx} = 1$

$du = dx$

$\int 2(u+5)\sqrt{u} du$

$= \int (2u+10) u^{\frac{1}{2}} du$

$= \int 2u^{\frac{3}{2}} + 10u^{\frac{1}{2}} du$ (3)

$= \frac{2}{\frac{5}{2}} u^{\frac{5}{2}} + \frac{10}{\frac{3}{2}} u^{\frac{3}{2}} + C$

$= \frac{4}{5} \sqrt{(x-5)^5} + \frac{20}{3} \sqrt{(x-5)^3} + C$

$$d) \frac{Bx + \alpha x + \alpha B}{\alpha Bx}$$

$$= \frac{-1}{2}$$

(2)

$$e) \tan^{-1}(-\sqrt{3}) = \frac{-\pi}{3} \quad (1)$$

$$f) P = \frac{-3(3) + 2(-2)}{-3+2}, \frac{-3(-6) + 2(4)}{-3+2}$$

$$P = (13, -26)$$

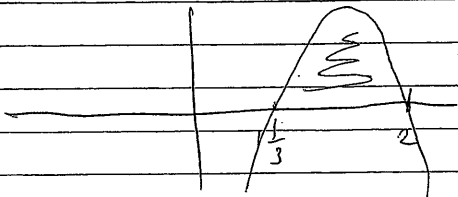
(2)

$$g) \frac{x}{2-x} - 2 \geq 0$$

$$\frac{2-x-2(2-x)}{2-x} \geq 0$$

$$(3x-4)(2-x) \geq 0$$

$$\frac{1}{3} \leq x \leq 2$$



Question 12.

$$a) \text{ Prove } 5^n > 20n - 1$$

I, Prove true when $n=3$

$$\text{LHS} = 5^3 = 125$$

$$\text{RHS} = 20 \times 3 - 1 = 59$$

$$\text{LHS} > \text{RHS}$$

$\therefore$ true when $n=3$.

II, Assume true for $n=k$.

$$5^k > 20k - 1$$

III, Prove true for $n=k+1$.

$$\text{RTP } 5^{k+1} > 20(k+1) - 1$$

i.e. Prove $\text{LHS} - \text{RHS} > 0$

$$\text{LHS} - \text{RHS} = 5^{k+1} - 20(k+1) - 1$$

$$= 5^k \cdot 5 - 20k - 20 - 1$$

$$> 5(20k-1) - 20k - 19$$

$$= 100k - 5 - 20k - 19$$

$$= 80k - 24$$

$$> 0 \text{ as when } k \geq 3, 80k - 24 \geq 216$$

IV, Blah.

$$b) i) P(\text{does not rain for a week}) = \left(\frac{1}{3}\right)^7 = \frac{1}{2187}$$

$$ii) P(SM) + P(MT) + P(TW) + P(WT) + P(TF) + P(FS)$$

$$= \left(\frac{2}{3}\right)^2 \times \left(\frac{1}{3}\right)^5 \times 6$$

$$= \frac{8}{729}$$

c) $u = \sin x$ when $x = \frac{\pi}{2}$, $u = \sin \frac{\pi}{2} = 1$
 $\frac{du}{dx} = \cos x$
 $du = \cos x dx$ when $x=0$, $u = \sin 0 = 0$

$$I = \int_0^1 u^2 du$$

$$= \left[\frac{u^3}{3} \right]_0^1$$

$$= \frac{1^3}{3} - \frac{0^3}{3}$$

$$= \frac{1}{3}$$

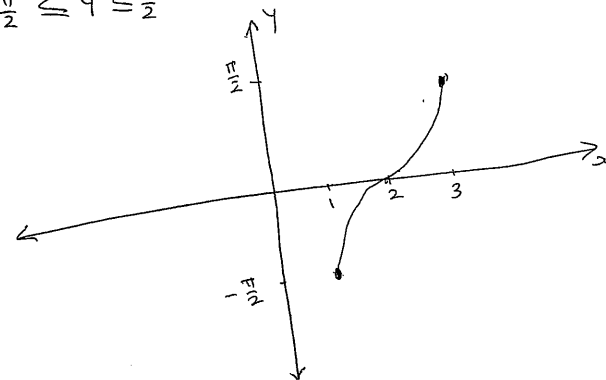
d) $\angle BAC = \angle BCD$ ($\angle$ in alt. segment)
 $= 50^\circ$
 In $\triangle BAF$, $\angle ABF = 40^\circ$ ($\angle$ sum $\triangle$)
 Join OC
 $\angle OCB = 90^\circ$ (tangent $\perp$ radius)
 $\therefore \angle OCB = 40^\circ$ (comp. $\angle$ s)
 As $OC = OB$ (radii)
 $\triangle OCB$ is isosceles
 $\therefore \angle OBC = 40^\circ$ (base $\angle$ s of isos $\triangle$)
 $\angle ABC = \angle ABF + \angle FBC$ (adj. $\angle$ s)
 $= 40^\circ + 40^\circ$
 $= 80^\circ$
 $\angle ACE = \angle ABC$ ($\angle$ in alt. segment)
 $= 80^\circ$

e) $y = \sin x$

$$D: -1 \leq x-2 \leq 1$$

$$1 \leq x \leq 3$$

$$R: -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



f) $f(x) = \sin x - \frac{x}{2}$

$$f'(x) = \cos x - \frac{1}{2}$$

$$f(2) = \sin(2) - \frac{2}{2}$$

$$= \sin(2) - 1$$

$$f'(2) = \cos(2) - \frac{1}{2}$$

$$= -0.91615 \dots$$

$$= -0.00907 \dots$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 2 - \frac{f(2)}{f'(2)}$$

$$= 2 - \frac{-0.00907 \dots}{-0.91615 \dots}$$

$$= 1.900995594 \dots$$

$$= 1.90 \text{ (2 dec. pl.)}$$

Question 13

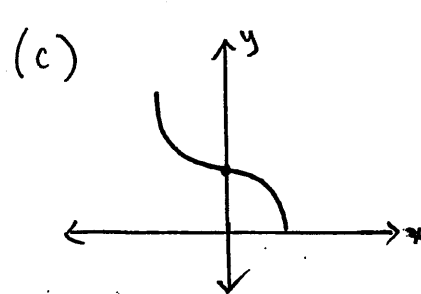
$$\begin{aligned}
 (a) \quad \int_0^2 \frac{dx}{4x^2 + 20} &= \frac{1}{4} \int_0^2 \frac{dx}{x^2 + 5} \\
 &= \frac{1}{4\sqrt{5}} \left[\tan^{-1} \frac{x}{\sqrt{5}} \right]_0^2 \\
 &= \frac{\tan^{-1} \left(\frac{2}{\sqrt{5}} \right)}{4\sqrt{5}}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad (i) \quad LHS &= \frac{dT}{dt} & T &= 20 + Ae^{-kt} \\
 &= -kAe^{-kt} \\
 &= -k(20 + Ae^{-kt} - 20) \\
 &= -k(T - 20) \\
 \therefore LHS &= RHS & \therefore T &= 20 + Ae^{-kt} \text{ is a soln. of the equation}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \text{When } t=0, T=100 \quad \text{when } t=15, T=70 \\
 100 &= 20 + A \Rightarrow A=80 \\
 70 &= 20 + 80e^{-15k} \\
 e^{-15k} &= \frac{5}{8} \Rightarrow k = -\frac{1}{15} \ln \left(\frac{5}{8} \right) \doteq 0.031
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad 25 &= 20 + 80e^{-0.031t} \\
 e^{-0.031t} &= \frac{5}{80} \Rightarrow t = \frac{-1}{0.031} \ln \left(\frac{5}{80} \right) \\
 \therefore t &\doteq 88.5 \text{ minutes}
 \end{aligned}$$

Question 13



$$\begin{aligned}
 V &= \pi \int_0^\pi x^2 dy \\
 &= \pi \int_0^\pi \cos^2 y dy \\
 &= \frac{\pi}{2} \int_0^\pi (\cos 2y + 1) dy \\
 &= \frac{\pi}{2} \left[\frac{\sin 2y}{2} + y \right]_0^\pi \\
 &= \frac{\pi}{2} [0 + \pi - (0 + 0)] \\
 \therefore V &= \frac{\pi^2}{2} \text{ units}^3
 \end{aligned}$$

$$(d) (i) P(x) = x^4 + 5x^3 + 4x^2 - 8x - 8$$

$$\begin{aligned}
 P(-1) &= 1 - 5 + 4 + 8 - 8 \\
 &= 0 \quad \therefore (x+1) \text{ is a factor}
 \end{aligned}$$

$$\begin{aligned}
 P(-2) &= 16 - 40 + 16 + 16 - 8 \\
 &= 0 \quad \therefore (x+2) \text{ is a factor} \\
 \text{Hence } (x+1)(x+2) &\text{ is a factor}
 \end{aligned}$$

$$(ii) \quad (x^2 + 3x + 2) Q(x) = x^4 + 5x^3 + 4x^2 - 8x - 8$$

$$\text{By inspection } Q(x) = x^2 + 2x - 4$$

Question 13

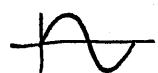
(e) $\tan \theta = \sin 2\theta$

$$\frac{\sin \theta}{\cos \theta} = 2 \sin \theta \cos \theta$$

$$2 \sin \theta \cos^2 \theta - \sin \theta = 0$$

$$\sin \theta (2 \cos^2 \theta - 1) = 0$$

$\sin \theta = 0$ or $\cos \theta = \pm \frac{1}{\sqrt{2}}$



(acute $\angle = \frac{\pi}{4}$)

$$\begin{array}{c|c} * & A \\ \hline S & C \\ \hline * & * \end{array}$$

$$\therefore \theta = 0, \pi, 2\pi, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Question 14

$$a) x = Vt \cos \theta \quad y = Vt \sin \theta - 5t^2$$

$$i) \text{When } x = 20, \\ y = 4 \\ t = 5$$

$$20 = 5V \cos \theta \quad 4 = 5V \sin \theta - 125 \\ 4 = V \cos \theta \quad \frac{129}{5} = V \sin \theta \quad (2) \\ V = \frac{4}{\cos \theta} \quad (1)$$

Sub (1) into (2)

$$iii) \text{Time of flight} = 2 \times 2.58 \\ = 5.16$$

$$\frac{129}{20} = \tan \theta$$

$$\text{Range} = 26.1 \times 5.16 \times \cos 81^\circ 11' \\ = 20.64$$

$$\theta = \tan^{-1} \left(\frac{129}{20} \right) \\ = 81^\circ 11'$$

$$V = \frac{4}{\cos 81^\circ 11'}$$

$$= 26.1 \text{ ms}^{-1}$$

ii) Max height when $y' = 0$

$$y' = V \sin \theta - 10t$$

$$V \sin \theta = 10t$$

$$t = \frac{V \sin \theta}{10}$$

$$= 2.58 \text{ s}$$

$$b) P(2ap, ap^2) \quad Q(2aq, aq^2)$$

$$i) M_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq} \\ = \frac{a(p+q)(p-q)}{2a(p-q)} \\ = \frac{p+q}{2}$$

Equation of PQ:

$$y - ap^2 = \frac{p+q}{2} (x - 2ap)$$

$$2y - 2ap^2 = (p+q)x - 2ap^2 - 2apq^2$$

$$(p+q)x - 2y - 2apq = 0$$

$$ii) x^2 = 4ay \\ y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

$$\text{when } x = 2ap$$

$$\frac{dy}{dx} = \frac{2ap}{2a}$$

$$= p$$

iii) If tangent at P is parallel to normal at Q then gradient of normal at Q is p and gradient of tangent at Q is $\frac{-1}{p}$.

$$\therefore q = \frac{-1}{p}$$

Substituting into PQ:

$$(p+q)x - 2y + 2a = 0$$

when $x = 0$

$$-2y + 2a = 0$$

$$y = a$$

$\therefore$ PQ passes through $(0, a)$

c)

i) $a = x - 2$

when $t=0$, $x=0$

$\therefore a = -2$

ii) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = x - 2$

$\frac{1}{2} v^2 = \int x - 2 \, dx$

$v^2 = 2 \int x - 2 \, dx$

$= \frac{2}{2} (x-2)^2 + C$

When $x=0$, $v=2$

$4 = 4 + C$

$C = 0$

$\therefore v^2 = (x-2)^2$

iii) when $x=0$, $v=2$

$\therefore v = -(x-2)$

$v = 2 - x$

$\frac{dx}{dt} = 2 - x$

$\frac{1}{2-x} dx = dt$

$t = -\ln(2-x) + C$

when $t=0$, $x=0$

$0 = -\ln 2 + C$

$C = \ln 2$

$t = -\ln(2-x) + \ln 2$

$\ln 2 - t = \ln(2-x)$

$x = 2 - e^{\ln 2 - t}$
 $= 2 - 2e^{-t}$